

OOPS(PBL) PROGRAM - 3

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Topic: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap.

INPUT :

```
// using binary search and quick sort
```

```
#include <iostream>
```

```
using namespace std;
```

```
void read(int[], int);
```

```
void display(int[], int);
```

```
void swap(int*, int*);
```

```
void search(int[], int);
```

```
void sort(int[], int, int);
```

```
int partition(int[], int, int);
```

```
void search(int[], int, int, int);
```

```
int main() {
```

```
    int a;
```

```
    int x;
```

```
    int y;
```

```
    int t;
```

```
    cout << "Enter number of elements in array: ";
```

```
    cin >> a;
```

```
    int arr[a];
```

```
    read(arr, a);
```

```
    display(arr, a);
```

```
    sort(arr,0,a-1);
```

```
    cout<< endl << "Sorted array is: ";
```

```
    for (int i = 0; i < a; i++) {
```

```
        cout << arr[i] << " ";
```

```
    }
```

```
cout << endl;
cout << "Enter Element to Find: ";
cin >> t;
search(arr,0,a-1,t);
```

```
    return 0;
}
```

```
void swap(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}
```

```
void read(int array[], int a) {
    cout << "Enter Array elements: \n";
    for (int i = 0; i < a; i++) {
        cin >> array[i];
    }
}
```

```
void display(int array[], int a) {
    cout<< endl << "Entered array is: ";
    for (int i = 0; i < a; i++) {
        cout << array[i] << " ";
    }
    cout << endl;
}
```

```
int partition(int array[], int l, int h){
    int pivot = array[h];
    int i = l-1;
    for(int j = l; j < h; j++){
```

```

        if(array[j]<pivot){
            i++;
            swap(&array[i],&array[j]);
        }
    }
    swap(&array[i+1],&array[h]);
    return (i+1);
}

```

```

void sort(int array[],int l, int h){
    if(l<h){
        int pi = partition(array,l,h);
        sort(array,l,pi-1);
        sort(array,pi+1,h);
    }
}

```

```

void search(int array[], int l, int h, int t){

```

```

    if (l > h) {
        cout << endl << "Not Found";
        return;
    }

```

```

    int mid = l + (h - l) / 2;

```

```

    if(t== array[mid]){
        cout << endl << "Found at index: "<< mid;
    }
    else if(array[mid] < t ){

```

```

        search(array,mid+1,h,t);
    }

```

```

    else if(array[mid] > t ){
        search(array,l,mid-1,t);
    }
}

```

```
else {  
    cout << "Not Found";  
}  
  
}
```

OUTPUT :

Output

Enter number of elements in array: 5

Enter Array elements:

2 3 21 4 1

Entered array is: 2 3 21 4 1

Sorted array is: 1 2 3 4 21

Enter Element to Find: 21

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=== Code Execution Successful ===